

Advancing Perceptual Training in Arts Education through Immersive Extended Reality Systems

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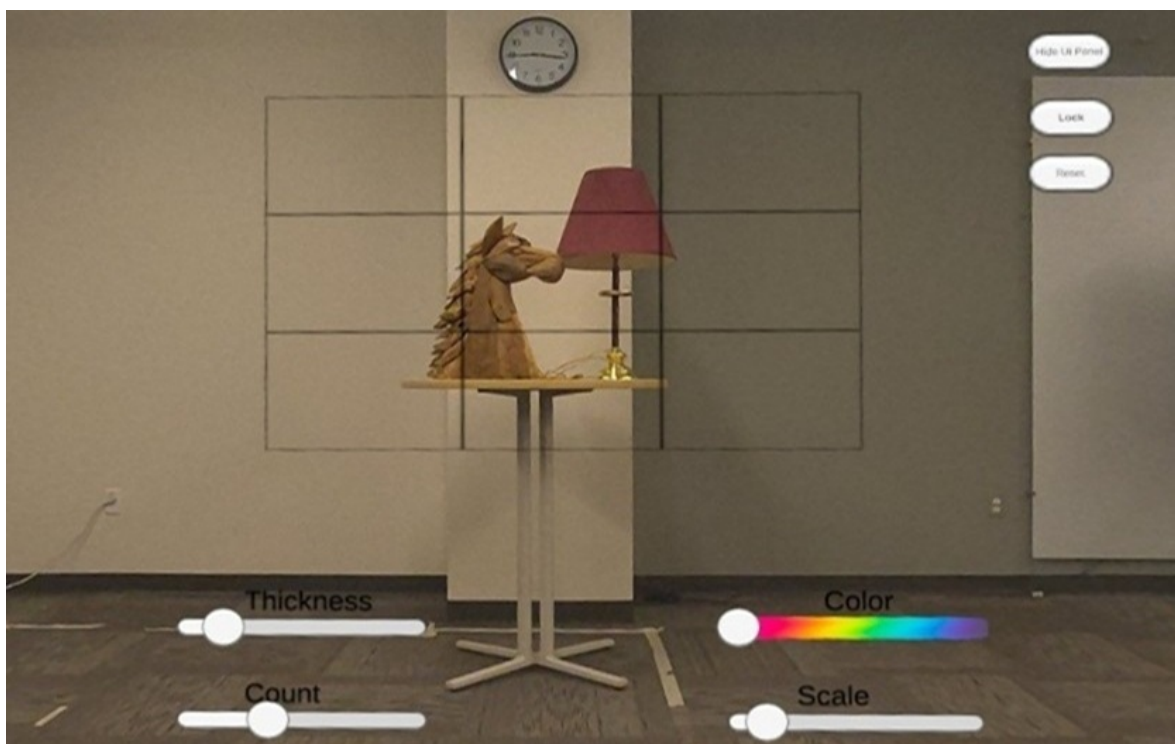


Figure 1: VR-based perceptual training system with grid overlay in passthrough mode on Meta Quest 3.

ABSTRACT

The accessibility of affordable virtual reality (VR) headsets with passthrough capabilities offers new opportunities in arts education. This paper introduces a VR-based perceptual training system for arts, designed using Unity and featuring a customizable 3D grid overlay in passthrough mode on a Meta Quest 3. A pilot study with 14 participants showed improved drawing accuracy and spatial understanding, as self reported by the participants, with high confidence levels ($M=8.1$) and low motion sickness ($M=3.1$). We detail our pedagogical principles, study results, and suggest future evaluation studies.

Index Terms: Virtual Reality (VR), Perceptual Training, Arts Education, Observational Drawing, Immersive Learning Environment, Human-centered computing

1 INTRODUCTION

Observational drawing involves accurately perceiving and translating three-dimensional forms onto two-dimensional surfaces. On the

other hand, symbolic representation relies on internalized mental models of objects. The conflict between these processes is linked to different areas of the brain that handle perception and symbolization. To effectively train observational drawing, it is essential to incorporate exercises that help bypass symbolic representation. This intricate process necessitates the mastery of various interconnected perceptual competencies, including spatial awareness, proportional reasoning, contour recognition, edge detection, negative space identification, and figure-ground distinction.

Traditional arts pedagogy has acknowledged that these perceptual skills can be systematically cultivated through structured training exercises. The seminal work of Betty Edwards, titled *Drawing on the Right Side of the Brain*[1], provides compelling evidence that drawing proficiency significantly improves when learners transition from symbolic thinking to visual-perceptual processing. Her methodologies, supported by subsequent research, have established perceptual training as a fundamental component of effective drawing instruction.

Existing interventions for perceptual training predominantly occur within conventional classroom environments using standard paper and art materials. Learners typically engage with traditional paper-based exercises, physical grid overlays, or printed reference materials. While these approaches have demonstrated effectiveness, they constrain engagement with depth perception, three-dimensional spatial relationships, and embodied spatial aware-

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ness—elements that are critical for translating observed three-dimensional forms into accurate two-dimensional representations.

The emergence of affordable virtual reality technology, particularly with high-quality passthrough capabilities, introduces a significant opportunity to enhance perceptual training using immersive three-dimensional environments. To the best of our knowledge, this initiative marks the first application of virtual reality technology specifically for perceptual training in the field of arts education. By capitalizing on virtual reality's capacity for spatial immersion, real-time depth cues, and the interactive manipulation of visual reference frameworks, it is possible to amplify the perceptual gains documented through traditional training methods while also introducing novel dimensions of spatial understanding.

This innovative approach holds particular promise for learners who experience challenges with drawing and visual representation. Many individuals find conventional drawing instruction difficult due to obstacles in effectively translating observed three-dimensional forms into accurate two-dimensional representations. The immersive three-dimensional grid overlay provides a structured perceptual scaffold that can render spatial relationships more explicit and accessible, thereby potentially mitigating barriers faced by learners who struggle with traditional drawing techniques.

2 RELATED WORK

The work titled *The Right Side of the Brain* (1989) [1] serves as a foundational text in the field of drawing instruction, positing that drawing skills can be cultivated by guiding individuals to shift from symbolic, analytical reasoning to a more visual-perceptual processing mode. Edwards' exercises, which include blind contour drawing, negative space recognition, and upside-down drawing, facilitate learners in transcending their internalized symbolic representations, thereby enhancing their ability to accurately observe visual information.

Empirical studies have provided substantial evidence in support of perceptual training approaches. *Salome* (1965) [7] presented early findings indicating that students engaged in perceptual training focused on contours and visual details produced drawings that were more representative and accurate compared to those receiving traditional instruction. Building upon this foundational research, *Speck* (1984) [8] demonstrated that eight-year-old students exhibited significant improvements in clarity, proportion, and spatial accuracy after participating in a six-week perceptual training program. *Richards* (1988) [6] extended these findings to a cross-cultural context, revealing enhanced drawing skills among both Jamaican and American students after undergoing perceptual training. More recently, *Mostert* (2022) [3] observed comparable benefits in Grade 5 learners, noting improved drawing performance, increased motivation, and enhanced artistic confidence.

Szeto and Salome (1977) [9] investigated the influence of perceptual training-drawing exercises on essential visual functions pertinent to drawing. Their study found that perceptual training improved the ability to process complex spatial relationships by focusing on "points of maximal contour," such as edges, curves, and directional shifts. These findings suggest that perceptual training can alleviate the tendency toward premature visual closure, thereby fostering a more accurate interpretation of visual stimuli. *Orde* (1997) [4] further posited that drawing should be perceived as a teachable skill rooted in visual-perceptual and spatial processing, as opposed to being considered solely an innate talent.

Recent research has explored technology-mediated approaches to support perceptual training in drawing. *Pyakurel et al.* (2023) [5] investigated how projected light cues can alter perception during observational drawing by helping users shift between distal mode (D-Mode) and proximal mode (P-Mode) perception. Their system projected interactive visual cues onto physical drawing objects to counteract constancy effects and guide visual attention. Similarly,

Laviole and Hachet (2012) [2] developed a spatial augmented reality system that projects 3D scenes onto paper sheets, allowing users to manipulate lighting and perspective before drawing. While these projection-based approaches demonstrate the potential of augmented reality to support perceptual training, they remain limited to 2D projection onto physical surfaces and suffer from occlusion problems when users' hands block the projected content. Projecting a 2D grid onto 3D surfaces aids in understanding 3D spatial relationships but doesn't aid in the crucial step of 3D to 2D paper projection necessary for perceptual drawing skills.

Beyond projection-based systems, the emergence of consumer virtual reality hardware has spawned numerous XR drawing applications, including *Tilt Brush*, *Quill*, *Gravity Sketch*, *OpenBrush*, and *Vermillion*. These applications primarily focus on enabling artistic expression through 3D sketching, painting in virtual space, and professional design workflows with features such as spatial brushstrokes, layering systems, and various brush effects. However, existing XR drawing tools do not address the fundamental perceptual training challenges identified in drawing education research, free-form expression but lack pedagogically grounded session, they lack pedagogically-grounded exercises for developing foundational observational skills. They approach rather than as a perceptual skill to be systematically developed, leaving users to rely on their existing perceptual habits without interventions that support transitions between the transition between different modes of visual perception or provide structured scaffolding for observational drawing training.

While prior studies have demonstrated that perceptual training enhances drawing efficiency and confidence, most interventions have remained confined to two-dimensional formats, such as paper-based classroom instruction. To date, no research has utilized virtual reality environments for perceptual training within arts education. Virtual reality technology offers distinct advantages that effectively address these limitations, including spatial immersion to engage depth perception, the provision of stable three-dimensional reference frameworks visible through passthrough mode, and embodied spatial exploration that may strengthen the correlation between visual perception and drawing practice.

3 VR-BASED TRAINING SYSTEM

This section presents the technical implementation and functionality of the VR-based perceptual training system. Section 3.1 describes the development framework and deployment platform, while Section 3.2 details the system's core features and customizable parameters designed to support spatial learning and perceptual skill development.

3.1 Technical Development

The perceptual training artifact is developed in Unity using the Meta XR SDK. It features a built-in three-dimensional grid overlay system to help users comprehend spatial dimensions. The artifact utilizes passthrough functionality, enabling users to see the real world while engaging in drawing activities. A user-friendly interface allows customization of the grid, giving users the flexibility to adjust it according to their preferences. After the system was completed, it was tested and deployed on the Meta Quest 3 to ensure functionality and enable effective interaction in real-world drawing scenarios.

3.2 System Description

The VR based training system provides the grid anchored in world space, remaining fixed within the user's physical surroundings irrespective of head movements. This design allows users to navigate around the grid and observe it from various perspectives, thereby enhancing depth perception, spatial orientation, and positional awareness.

The system incorporates customizable parameters to accommodate a range of perceptual tasks, including:

- **Grid thickness control** for improved clarity and visibility
- **Color customization** to enhance contrast across different environments
- **Grid density adjustment** allowing modification of grid line numbers
- **Grid resizing** for different perceptual tasks
- **Dynamic grid positioning** for context-specific placement

The structured visual reference serves as an effective tool for facilitating edge alignment, proportional estimation, and contour recognition. By encouraging users to align objects within their real-world environment to the geometric constraints of the grid, the system promotes enhanced visual discrimination and improved focus on contours, negative spaces, and angular relationships.

4 PILOT STUDY

An initial pilot study was conducted to validate the VR-based perceptual training system with 14 participants. Section 4.1 details the experimental approach and assessment framework, Section 4.2 presents numerical results from the 10-point Likert scale questionnaire, Section 4.3 highlights participant observations and feedback, and Section 4.4 examines the significance of these findings for system refinement and subsequent research.

4.1 Methodology

The objective of this study was to evaluate the system's usability, assess user comfort, enhance drawing accuracy, and gather feedback for interface improvement. The research involved 14 participants (N=14) representing a spectrum of prior VR experience.

After a brief orientation to the VR system and its controls, participants engaged in drawing tasks utilizing the grid features for sessions ranging from 7 to 15 minutes. They were encouraged to experiment with various customization features of the grid—including color, thickness, scale, and spatial placement adjustments—to investigate how different configurations could facilitate their drawing process.

A comprehensive post-experience survey, employing a 10-point Likert scale (1 = Strongly Disagree, 10 = Strongly Agree), was administered. This survey consisted of 14 quantitative items organized into four categories:

- Comfort and Physical Experience (4 items)
- User Interface Usability (5 items)
- Drawing Effectiveness (2 items)
- Overall System Usability (3 items)

Additionally, open-ended questions were included to gather qualitative feedback. **Figure 2** demonstrates the snapshots taken from the pilot study conducted with the participants.

4.2 Results

The pilot study yielded encouraging results across all evaluated dimensions, with mean scores ranging from 6.4 to 8.1 on the 10-point scale. **Table 1** below presents comprehensive results.

Key findings from the analysis are as follows:

- **Drawing Effectiveness**—The grid functionality received the highest rating (M=7.9, SD=1.7), confirming that three-dimensional grid overlays significantly enhance drawing accuracy;
- **User Confidence**—A notably high confidence level (M=8.1, SD=1.6) suggests an intuitive system design;
- **Independent Task Completion**—High ratings (M=7.9, SD=1.8) reflect successful usability;

Table 1: Pilot Study Results (N=14)

Category & Item	Mean	SD
Comfort		
Overall comfort	7.4	1.8
Minimal Motion Sickness*	3.1	1.9
Weight Acceptable*	3.9	2.3
Sustained Comfort	6.9	2.6
UI Usability		
Ease of UI features	7.1	2.1
Interaction accessibility	7.3	2.0
Grid Adjustment	6.4	2.4
Intuitive Layout	7.1	2.2
System Responsiveness	6.9	2.5
Drawing		
Grid Improved Accuracy	7.9	1.7
Customization Value	7.1	2.0
Overall		
System Usability	7.6	1.9
Independent Completion	7.9	1.8
User Confidence	8.1	1.6

*Lower scores indicate positive outcomes

- **Physical Comfort**—A low incidence of motion sickness (M=3.1, SD=1.9) indicates that the virtual reality interaction design is appropriate;
- **Interface Usability**—Most metrics surpassed the threshold of 7.0, although the grid adjustment controls received a rating of (M=6.4, SD=2.4), suggesting areas for improvement.

4.3 Qualitative Feedback

The findings highlight that many participants found the grid to be beneficial in enhancing their ability to perceive proportions with greater clarity and to understand the spatial relationships between objects. Participants self-reported improvements in drawing accuracy and spatial understanding when using the grid system. Several participants remarked that the grid facilitated a focus on observational accuracy rather than relying on symbolic representations. As one participant noted, "the grid prompted me to observe what was actually present instead of what I assumed should be there." This observation exemplifies the perceptual transition from symbolic thinking to visual-perceptual processing, which the system is specifically designed to promote.

4.4 Discussion

The highest rating for grid-enhanced accuracy (M=7.9) substantiates our design hypothesis and provides empirical evidence for the extension of perceptual training into immersive environments. While these improvements are self-reported by participants, they provide valuable preliminary insights into user experience. Notably, strong user confidence (M=8.1) and independent task completion (M=7.9) indicate that the system facilitates creative work while maintaining manageable cognitive load. Additionally, the minimal incidence of motion sickness suggests that passthrough-based drawing applications can be utilized comfortably over extended periods.

Given that this pilot study involved 14 participants in single-session assessments with self-reported metrics, the findings offer preliminary validation suitable for early-stage development. The favorable outcomes across all metrics warrant further exploration through more extensive studies with objective performance measures.



Figure 2: Pilot Study conducted with 14 participants

5 PLANNED EVALUATION STUDIES

Building upon pilot study insights, a comprehensive two-phase evaluation will assess the refined VR system's usability and educational effectiveness. Section 5.1 describes the usability testing protocol with 25 participants to evaluate interaction quality and system performance, while Section 5.2 outlines a comparative study with 30 students to measure the system's impact on perceptual skill development against traditional classroom instruction.

5.1 Phase One—Usability Testing

A total of twenty-five participants will participate in usability testing. Upon completion of a pre-session questionnaire, participants will engage with the virtual reality (VR) grid system for a duration of 20 minutes, during which they will explore the grid customization features and dynamic positioning capabilities. Following the session, participant feedback will be collected to assess the ease of interaction, comfort level, visual clarity, usability of controls, and effectiveness in supporting perceptual focus.

5.2 Phase Two—Comparative Study

A comparative study will be conducted involving 30 learners, who will be divided equally into two groups. Group A will partake in traditional classroom-based perceptual exercises, while Group B will utilize the VR-based training system. Both groups will complete structured perceptual tasks specifically designed to enhance the accuracy of representational drawing, including:

- **Edge tracing** to enhance contour recognition
- **Negative space drawing** to identify shapes around and between objects
- **Proportion estimation** to improve size and distance judgment
- **Perspective alignment** for understanding horizons and vanishing points
- **Figure-ground distinction** to separate forms from backgrounds

The efficacy of the immersive training system in enhancing perceptual and representational skills is the focus of this research. Statistical analyses, including Cohen's d and an Intraclass Correlation Coefficient (ICC) > 0.80 for reliability, are anticipated to demonstrate significant improvements in learning outcomes.

6 CONCLUSION AND FUTURE WORK

The combination of immersive passthrough visualization with customizable grid controls enables a personalized training experience for learners while reinforcing key perceptual concepts through embodied interaction. The findings from the pilot study affirm the system's usability and illustrate its potential effectiveness in enhancing representational accuracy and cognitive visual skills.

The principal contribution of this work is the advancement of traditional two-dimensional perceptual training methodologies into a dynamic three-dimensional environment. This provides learners with improved spatial awareness through the use of customizable visual scaffolding. In contrast to conventional classroom-based teaching methods, this system offers real-time, adjustable geometric references that learners can contextually position within their physical surroundings.

Future research will concentrate on broadening the system's pedagogical functionalities through several initiatives:

- **Enhanced interface controls:** Developing more precise and intuitive manipulation controls for fine-tuning grid placement, orientation, and configuration parameters to further improve the user experience;
- **Upside-down transformation:** Implementing an upside-down transformation to aid learners in focusing on shapes and spatial relationships while minimizing symbolic biases;
- **Memory training modules:** Developing memory training modules to enhance visual retention;
- **Guided perceptual exercises:** Providing guided perceptual exercises with real-time instructional prompts; and
- **Cross-platform XR compatibility:** Establishing cross-platform extended reality (XR) compatibility to increase accessibility.

These enhancements are designed to advance the system into a comprehensive immersive training framework that fosters the development of perceptual and cognitive skills specifically within the domain of arts education.

REFERENCES

- [1] B. Edwards. *Drawing On the Right Side of the Brain: A Course in Enhancing Creativity and Artistic Confidence*. Rev. ed. J.P. Tarcher, 1989. 1, 2
- [2] J. Laviolle and M. Hachet. Papart: interactive 3d graphics and multi-touch augmented paper for artistic creation. In *2012 IEEE symposium on 3D user interfaces (3DUI)*, pp. 3–6. IEEE, 2012. 2
- [3] W. A. Mostert. The construction of knowledge through visual perceptual training in visual arts. *South African Journal of Childhood Education*, 2022. 2
- [4] B. J. Orde. Drawing as visual-perceptual and spatial ability training. *National Convention of the Association for Educational Communications and Technology*, 1977. 2
- [5] B. Pyakurel, S. Endow, C. Donjuan, and C. Torres. Cogcues: Shifting perception through interactive projected cues in still life drawing. In *Proceedings of the 15th Conference on Creativity and Cognition*, pp. 171–184, 2023. 2
- [6] A. G. Richards. Perceptual training in drawing among students from two countries. *Studies in Art Education*, 29(3):302–308, 1988. 2
- [7] R. A. Salome. The effects of perceptual training upon the two-dimensional drawings of children. *Studies in Art Education*, 7(1):18–33, 1965. 2
- [8] C. Speck. A perceptual training and drawing programme to assist the drawing development of eight-year-old children. Master's thesis, Canberra College of Advanced Education, 1984. 2
- [9] J. W. Szeto and R. A. Salome. The effects of scanning practice and perceptual training upon several visual functions: Implications for drawing. *Studies in Art Education*, 19(1):45–51, 1977. 2